

TABLE OF CONTENTS

E-401 AIR INSULATED BUS-WORK

	Page No.
1. SCOPE OF WORK.....	E-401- 2
2. CODES AND STANDARDS	E-401- 2
2.1 Indonesian National Standards.....	E-401- 2
2.2 International Electrotechnical Commission	E-401- 2
3. OPERATING DESIGN CONDITIONS	E-401- 3
4. DETAILED REQUIREMENTS	E-401- 3
4.1 General	E-401- 3
4.2 Ratings	E-401- 3
4.3 Aerial Wire Conductor.....	E-401- 4
4.4 Tubing Conductors and Tubing Connectors	E-401- 5
4.5 String or Tension Insulators.....	E-401- 5
4.6 Post Insulators.....	E-401- 5
4.7 Clamps and Fittings	E-401- 6
4.8 Corona.....	E-401- 6
4.9 Generator Connection	E-401- 7
5. PERFORMANCE TESTING AND GUARANTEES.....	E-401- 7
6. QUALITY ASSURANCE/QUALITY CONTROL	E-401- 7

E 401 AIR INSULATED BUS WORK

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation (including integration to the exiting buses), commissioning and testing of air insulated bus work in accordance with Part 4, Subsection 1.2 Summary of Work, and refer to the drawings

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards and the following specified standards:

2.1 Indonesian National Standards

- PUIL : Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electric Installation-Indonesia)
- SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

- IEC 60060 : High-voltage test techniques
- IEC 60105 : Recommendation for commercial-purity aluminum busbar material
- IEC 60120 : Dimensions for ball and socket couplings of string insulator units
- IEC 60121 : Recommendation commercial annealed aluminium electrical conductor wire
- IEC 60168 : Tests on indoor and outdoor post insulators of ceramic material or glass for systems with nominal voltages greater than 1000V
- IEC 60273 : Characteristic of indoor and outdoor post insulators for systems with nominal voltages greater than 1000V
- IEC 60305 : Insulators for overhead lines with a nominal voltage above 1000V- Ceramic or glass insulator units for a.c. systems - Characteristics of insulator units of the cap and pin type
- IEC 60372 : Locking devices for ball and socket couplings of string insulator units - Dimensions and tests
- IEC 60383 : Insulators for overhead lines with a nominal voltage above 1000V
- IEC 60437 : Radio interference test on high-voltage insulators
- IEC 60471 : Dimension of clevis and tongue couplings of string insulator units
- IEC 60575 : Thermal-mechanical test and mechanical performance test on string insulator units

- IEC 60672 : Ceramic and glass insulating materials
- IEC 60889 : Hard-drawn aluminium wire for overhead line conductors
- IEC 61089 : Round wire concentric lay overhead electrical stranded conductors.

3. OPERATING DESIGN CONDITIONS

The 500 kV conventional substations shall be designed for outdoor operation in tropical climate along the seacoast. Refer to Part 4, Sections 1.1, 2.14 and 2.16 for meteorological, seismic and environmental data of the site.

4. DETAILED REQUIREMENTS

4.1 General

Air insulated bus work shall be designed for conductor temperature not to exceed 75°C, considering solar impingement and no wind based on maximum temperatures expected, and minimum temperature without any external or current thermal input.

Safety factors for design shall be not less than 50% of the elastic limit for metal, and 50% of the electromechanical test for porcelain material.

4.2 Ratings

(1)	Nominal voltage level	500 kV	Neutral
(2)	Maximum voltage level	525 kV	36 kV
(3)	Frequency	50 Hz	50 Hz
(4)	Continuous current level	3,150 A	2000 A
(5)	Momentary current level	100 kA rms	100 kA rms
(6)	Short time (1 second) current	40 kA	40 kA
(7)	Phase-to-ground insulation:		
	a. Impulse withstand voltage (1.2 x 50 μ s crest)	1800 kV	170 kV
	b. 50 Hz dry (and wet) withstand test voltage.....	780 kV	70 kV
	c. Switching impulse withstand test voltage	1,175 kV	N/A
(8)	Radio Influence Voltage (RIV)		
	The maximum RIV at 1 MHz, shall be :		
	a. Applied phase-to-ground voltage.....	335 kV	
	b. RIV microvolt	2,500 μ V	N/A

Measurements shall be in accordance with NEMA 107 procedure. In marking the RIV test, radio noise readings shall be taken at various points to determine the break point.

- (9) Creepage level to ground over insulation.....16,275 mm 1,116 mm
- (10) Clearance distances (minimum)
 - a. Phase-to-phase:
 - a.1 Rigid bus spacing.....7.7 m
 - a.2 Strain bus spacing.....9.0 m
 - a.3 Minimum clearance5.2 m
 - b. Phase-to-ground:
 - b.1 Minimum4.1 m
 - b.2 Minimum live parts height above ground7.5 m
 - b.3 Shield wire over conductor7.5 m

4.3 Aerial Wire Conductor

Aerial wire conductor shall normally be made of aluminium, but where impractical may utilize aluminium-clad steel core with aluminium conductors overlay. Based on voltage level, use of bundle conductors to minimize corona, radio influence voltage, and noise may be required.

Aerial conductor wire shall meet the requirements of IEC 61089 or equivalent.

Aerial ground wire shall be made of high strength stranded aluminum-clad steel, sized for lightning & fault return currents and equivalent to transmission line shield wire size at 12.7 mm diameter, seven (7) strands, with breaking strength at 83.6 Newton.

Particular care shall be taken at all times to ensure that the cables do not become kinked, twisted, flattened, or a braided in any manner. If cables or associated materials are damaged due to the Contractor's negligence or faulty equipment, the Contractor shall replace the damaged sections, including the furnishing of the necessary material at no additional cost. All sections of cable that are damaged by the applications of grips shall be replaced.

Appropriate initial stringing sag data shall be supplied. No prestressing of conductor will be required or allowed. No minus tolerances from the sag data will be allowed and the plus tolerances shall not exceed 120 mm.

No splicing of phase conductors is allowed.

The overhead shield wires shall be strung before the conductors are strung. Care shall be taken to ensure that the cable wire shall not touch the ground at any time. Appropriate stringing sag data shall be supplied. No minus tolerances from the sag data will be allowed and the plus tolerances shall not exceed six (6) inches.

4.4 Tubing Conductors and Tubing Connectors

Tubing shall be of hard drawn aluminum in accordance with IEC 60889 and shall have sufficient diameter and cross section to meet the mechanical and thermal requirements based on the maximum ambient temperature.

The tubing fittings and connectors shall be made of hard-drawn aluminium of sufficient size to minimize corona and radio influence voltage. Fittings shall be designed with rigid, slip, or expansion design to eliminate thermal expansion forces from being applied to insulators and equipment bushings.

Tubing shall be sized and supported such that maximum sag of the tubing span shall be less than one-half the tubing diameter. Tubing shall be provided with damping arrangements to eliminate natural frequency resonance due to wind.

4.5 String or Tension Insulators

String insulators shall meet the requirements of IEC 60383 and the mechanical characteristics shall meet the requirements of IEC 60305. The ball and socket couplings and other metallic couplings shall meet the requirements of IEC 60120 and IEC 60372 using corrosion-resistant, resilient metals of suitable mechanical strength.

All insulators shall be made of porcelain with chocolate brown glaze.

4.6 Post Insulators

Post insulators shall be designed and manufactured such that thermal and mechanical stresses shall not cause defects to develop.

Post insulators shall be dimensioned in accordance with IEC 60273 and type and routine tests applied as per IEC 60168. They shall be fully interchangeable units with solid core and cylindrical in shape, such that each unit can be used in either upright or inverted position. Mechanically strength graded units are acceptable, provided they meet the interchangeability requirements.

All post insulators shall be made of porcelain with chocolate brown glaze.

All post insulators shall be designed such that the forces of combined short circuit as per Clause 4.2 and wind loading of 150 km per hour shall not exceed 50% of the published strength of the insulators.

4.7 Clamps and Fittings

Clamps shall be designed to avoid deformation or separation of conductor stranding, including providing bell-shaped entry areas to eliminate kinking of conductor as it leaves the clamp.

All ferrous metal clamps, fittings, bolts, and other hardware shall be galvanized.

Suspension clamp shall be mechanically designed such that conductor will slip to relieve tension prior to failure of clamps. All ferrous metal suspension clamp used on aluminium-stranded conductors shall have soft pure aluminium liners for conductor protection. All suspension clamps shall allow free pivoting around the axis perpendicular to the plane formed by the conductor and insulator string.

Tension clamps shall be of the compression type such that mechanical efficiency is not compromised by external erection activities. Tension clamp assemblies shall be arranged to provide minimum clearance of 150 mm distance between jumper conductor and the rim of the insulator units.

All split pins for securing of fittings shall be phosphor bronze and backed with washers.

Spacer shall be installed immediately after clamping-in operations are completed in each pull, to minimize conductor damage due to contacts between the phase conductors. The equipment and methods used to install spacers shall be such that the conductors will not be damaged.

All bolted and compression connectors and equipment terminals which join aluminum to aluminium and aluminium to copper shall have agreed upon oxide inhibitor such as “NO-OX-ID A Specials” compound liberally applied to such contact surfaces in accordance with instructions. All bolts on aluminum connectors shall be installed with torque wrenches to recommended torque values.

4.8 Corona

Surfaces in contact with live potential shall contain no projection points or irregularities and they

shall be designed for a minimum of visible corona or brush discharge when the individual assemblies are completely assembled.

No visual corona shall exist at the rated maximum applied voltage. Lights shall be turned off in the test bay, and visual start and stop corona voltages shall be recorded with the corresponding radio noise voltages on both ascending and descending voltage values. Photographs shall be taken for each recorded voltage value. NEMA 107 test procedures shall be used.

4.9 Generator Connection

The generator line connection to the generator transformer shall meet all of the requirements outlined in this Technical Specification except where specifically modified in this Clause 4.9.

Suspension and tension insulator design shall have parallel strain paths between the conductor and tower such that breaking of one (1) insulator string will not result in conductor dropping to ground under normal loading.

Conductor shall be the four-bundle design per phase, using Hawk ACSR) aluminium clad steel core design. The tension in the conductor shall not exceed 1,360 kg under any design condition.

The following clearance and spacing shall be required for the 500 kV generator lines:

Phase-to-phase	9 m
Clearances to ground for safety over power plant roads	15 m
Clearances to ground for safety over plant road (for 17 m gantry crane)	25 m
Balances of clearances shall be as per Clause 4.2 of this Specifications.	

5. PERFORMANCE TESTING AND GUARANTEES

Insulators, fittings and bus work conductor assemblies shall be tested in accordance with IEC 60168, IEC 60305, IEC 60383 and IEC 60575.

A radio interference test on one (1) of each complete assembly style of insulators, including clamps, fitting and representative conductor length, shall be performed as per this Specification and NEMA 107 along with corona test.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

The quality assurance/quality control shall conform to the requirements of form included in Part 4 Subsection 2.7. The procedures specified shall include, but not be limited to, the following:

- (1) All test specified in Clause 5 of this Specification.

- (2) All special type tests performed for the specification as per Clause 5.

The quality assurance/quality control documentation as specified above shall include, but not be limited to, the following:

- (1) Shop Test Report.
- (2) Material Test Report.
- (3) All other documentation required by applicable codes and standards.

(Last page of E-401)